MILITARY SPECIFICATION

TANKS, HOT-DIP, DIRECTLY AND INDIRECTLY HEATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrically heated tanks for heating and maintaining selected working temperatures of preservatives, strippable coatings, and sealing compounds with equipment and accessories specified herein.

1.2 Classification. This specification covers hot-dip tanks of the following types and classes. The types and classes to be furnished shall be as specified (see 6.2.1).

Type I - Directly heated - External heater type
  Class A - 100 to 225° F heating range
  Class B - 100 to 400° F heating range
  Class C - 100 to 600° F heating

Type II - Directly heated - Immersion heater type
  Class A - 100 to 300° F heating range

Type III - Indirectly heated
  Class A - 100 to 300° F heating range
  Class B - 100 to 500° F heating range
  Class C - 150 to 600° F heating range

1.2.1 Size. The size of the tank shall be as specified (see 6.2.1). The minimum size of the tank shall be not less than 12 inches in length by 6 inches in width by 6 inches in depth. Size changes shall increase in increments of 6 inches in length, width, or depth as required. Dimensions specified shall be clear inside dimensions of the dip compartment and the dimensions shall be not less than specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motor, Alternating-Current, and Integral Horsepower (500 H.P. and Smaller).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner.

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MIL-C-104	- Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Method of.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-52950	- Crate, Wood, Open and Closed.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

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2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA ICS 1 - Industrial Controls and Systems.

ANSI/NEMA - ICS 6 Enclosures for Industrial Controls and Systems (Type 12).

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 3.3).

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3.2 Design. The hot-dip tank shall be new and one of the manufacturer's current models capable of heating and maintaining selected working temperature of preservative strippable coatings and sealing compounds in accordance with the requirements herein. The tanks shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Design characteristics shall be in accordance with the types specified in 3.2.1, 3.2.2, and 3.2.3.

3.2.1 Type I. Type I directly heated tank shall have the heating unit attached to the outside of the dipping compartment.

3.2.2 Type II. The type II tank shall be directly immersion heated. The heating unit shall be an immersion type heater, located in the bottom of the dipping compartment.

3.2.3 Type III. The type III indirectly heated tank shall have a dipping compartment submerged in a heat transfer fluid with the heating unit attached to or immersed in the heat transfer fluid container.

3.2.4 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the hot-dip tank. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.5 Material. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall not be used in contact with each other unless protected against galvanic corrosion.

3.2.5.1 Reclaimed materials. The hot-dip tank may contain reclaimed materials provided such materials will not jeopardize the hot-dip tank's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the hot-dip tank.

3.2.6 Energy efficiency. The hot-dip tank and its applicable components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the latest developments available with the industry.

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3.2.7 Mercury restriction. The hot-dip tank shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the hot-dip tank.

3.2.9 Environmental protection. The hot-dip tank shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the hot-dip tank shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.10 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.11 Safety devices. Each hot-dip tank shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. All parts subjected to high temperature shall be insulated, fully enclosed, or guarded to preclude injury to operating personnel. Type III, class C indirectly heated tanks shall be provided with means for preventing heat transfer fluid temperature in excess of 625°F. Except as otherwise specified by the procuring activity, the tank shall comply with the standards promulgated under OSHA 2206 applicable to the tank itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

3.2.12 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The hot-dip tank shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the hot-dip tank to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the hot-dip tank shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the hot-dip tank and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Electrical requirements. The hot-dip tanks shall be furnished complete with all electric motors, controls, relays, signal lights and switches, prewired where possible for operation when connected to source of power. All electrical equipment and the installation thereof shall be in compliance with ANSI/NEMA ICS 1 and enclosures shall be in compliance with ANSI/NEMA ICS 6, Type 12. Heating elements not specifically covered in the standard shall be covered and insulated to prevent contact with flammable liquids and vapors.

3.4.1 Power supply. Voltage and current characteristics of the tanks shall be as specified (see 6.2.1).

3.4.2 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with means to prevent loss of tightness.

3.4.3 Dielectric strength. All conducting components of the electrical circuit shall be capable of withstanding without damage twice the rated voltage plus 1000 volts root mean square (rms) between the circuit components and the dip-tank cabinet for a period of 1 minute at a frequency not less than the rated frequency of the power supply.

3.5 Performance.

3.5.1 Class A. Class A tanks with dip compartment capacities up to 10 cubic feet, when filled to 80 percent of capacity with a heat-transfer fluid, shall be capable of raising the temperature of the fluid with the cover closed, from 70° F ±5° F to not less than 210° F in not more than 3 hours and tanks with capacities from 10 to 20 cubic feet in not more than 4 hours. The fluid used for the heating requirements shall have an average specific heat content of not less than 0.50 Btu/lb. Beating requirements of tanks with dipping compartment capacities greater than 20 cubic feet shall be as specified (see 6.2.1).

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3.5.2 Class B and C. Class B and class C tanks with dip compartment capacities up to 10 cubic feet, when filled to 80 percent of capacity with a heat transfer fluid shall be capable of raising the temperature of the fluid, with the cover closed, from 70 + 5° F to not less than 375° F in not more than 4 hours, and tanks with capacities from 10 to 20 cubic feet in not more than 5 hours. The fluid used for the heating requirements shall have an average specific heat content of not less than 0.50 BTU/lb. Heating requirements of tanks with dipping compartment capacities greater than 20 cubic feet shall be as specified (see 6.2.1).

3.5.3 Operating controls and indicators. Operation of the hot-dip tank shall be controlled by a master switch, a programming switch, and a thermostatic switch. The thermostatic switch shall be of the indicating type or a thermometer shall be provided to indicate the temperature of the dip material. Indirectly heated tanks shall be provided with thermometer to indicate temperature of the heat transfer fluid. Two indicating lights shall be provided. One light shall function when the programmer has made current available to the thermostat and the other light shall function when current is being supplied to the heating units.

3.6 Components. The hot-dip tank shall consist of, but not be limited to, the following principle components.

3.6.1 Cabinet. The cabinet shall be a welded fabrication of sheet steel and shall rigidly support all component parts and a load in the dip compartment of not less than 125 pounds per cubic foot of dip compartment capacity. Unless otherwise specified (see 6.2.1), the cabinet shall have not less than 4 legs with bearing plates drilled for floor mounting, or have base supports to raise the tank not less than 3 inches above the floor. Supports shall be located crosswise and not greater than 40 inches apart. Unless otherwise specified (see 6.2.1), the dip compartment top shall be 36 +1 inch above floor level except for tanks requiring greater height for reason of capacity requirements. The cabinet construction shall be free of obstructions to the work area above the dip compartment. Controls and indicators shall be grouped, readily accessible for adjustment and maintenance, guarded to prevent contact with the dip material, and securely attached. Cabinet construction shall prevent contact between insulation and dip material. The cabinet shall provide access for inspection and replacement of heating units. A steel shelf or shield shall be provided above the controls and electrical equipment to provide protection for oils and waxes.

3.6.2 Dip compartment, The dip compartment of tanks with capacities up to 5 cubic feet shall be fabricated of sheet steel not less than 12 gauge, from 5 to 10 cubic feet not less than 10 gauge, and tanks with capacities greater than 10 cubic feet not less than 3/16-inch thick sheet steel. Temperature sensing elements of controls and indicators shall be recessed in the dip compartment side or bottom. There shall be no obstructions within the dip compartment. A drain shall be provided in the lowest part of the dip compartment. The drain for tanks up to 5 cubic feet shall be not less than 3/4 inch,

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from 5 to 10 cubic feet not less than 1 inch and larger than 10 cubic feet 1-1/2 inch nominal pipe size. The drain shall embody a valve and permit attachment of standard threaded pipe fittings. Type I directly heated tanks shall have heating units attached to the bottom of the dip compartment and no less than two sides. The side mounted heating units shall be located in the area between the compartment bottom and 2 inches below the dip material level with the dip compartment filled to 75 percent of capacity. When specified, tanks with dip compartment capacities of 150 gallons or less shall be provided with an overflow drain. The location of the overflow in the tank and overflow pipe size shall be as specified (see 6.2.1). The overflow requirements of tanks having dip compartment capacities greater than 150 gallons shall conform to the requirements of OSHA. There shall be no permanent deformation of the tank or leakage when the tank is filled with applicable material. The tank shall have a fill line to denote the maximum no load liquid or solution.

3.6.3 Heat transfer fluid jacket. Type III hot-dip tanks shall have a heat transfer fluid jacket fabricated of sheet steel of the thickness of not less than specified for the dip compartment (see 3.6.2). All seams shall be secured and there shall be no permanent deformation of the jacket or leakage of heat transfer fluid from the jacket when filled to its maximum working level. The jacket shall permit uniform heating of the bottom and four sides of the dip compartment to a height of 1 inch above the dip material upper working level. The jacket shall be provided with a filler opening and cap not less than 1-inch nominal pipe size. The jacket shall be vented to preclude pressure buildup due to heat. A drain not less than 3/4-inch nominal pipe size shall be provided at the lowest part of the jacket. The drain shall embody a valve and permit the attachment of standard threaded pipe fittings. Construction of the jacket shall prevent contact between heat transfer fluid, dip material, and insulation during filling, draining, normal operations, and shipment. The heating units shall be located in the lower two-thirds of the area between the jacket bottom and heat transfer fluid level.

3.6.4 Heating units. The heating units shall be of sufficient capacity to conform to the requirements of 3.5. The units shall be electrically insulated from the tank and shall be readily accessible for servicing or replacement without use of special tools. A removable cover or covers with gasket and conduit connector shall be fitted over the heating units to prevent entrance of flammable liquids and vapors. The cover shall be insulated to prevent heat loss.

3.6.5 Insulation. Each hot-dip tank shall be completely insulated on all four sides and the bottom with not less than three inches of nonsettling fiberglass or rockwool material. The tank outside jacket covering the insulation shall be fabricated of sheet steel of not less than 14 gauge.

3.6.6 Cover, removable. Unless otherwise specified (see 6.2.1), the tank shall be equipped with an insulated removable cover which shall fit the inside dimensions of the tank and rest on a suitable heat resisting gasket securely attached to the cover. The cover shall be equipped with suitable handles.

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3.6.6.1 Cover, automatic closing. When specified (see 6.2.1), the hot-dip. tank shall be provided with an automatic closing insulated cover hinged to the tank body to close over the dip compartment opening. The cover shall be provided with a handle for raising and lowering and shall embody a latch to hold the cover in the open position. The automatic closing shall be accomplished with a temperature-operated mechanism positioned above the compound level to ensure closing of the cover when excessive temperature exists in the ambient air. The cover, if open, shall close automatically when the temperature at the sensing element exceeds 250+ 30° F for class A tanks, and 360° F for class B and C tanks. The effort required to raise or close the cover and components attached thereto shall not exceed 25 pounds force. A counterbalance mechanism may be used when necessary to limit the effort. required to open and close the cover to not more than the 25 pounds force.

3.6.7 Master switch. Each hot-dip tank shall be provided with a master switch embodying a circuit breaker disconnect. The switch shall be capable of isolating from the power source all electrical equipment except the programming switch motor.

3.6.8 Programming switch. When specified (see 6.2.1), each hot-dip tank shall be provided with a programming switch of 7 day cycle. The programmer shall permit 7 day preelection of variable daily duty demands of 1 to 23 hours with provision of skipping days where operation is not required. Time indicating scales shall be calibrated in increments not greater than 1 hour.

3.6.9 Thermostatic switch. A thermostatic switch shall be provided for selection and automatic control of the dip material temperature. The thermostat shall permit selection of any temperature between not more than 100 and 225° F for type I, class A tanks, 100 to 300° F for type II and type III class A tanks, 100 to 500° F for class B tanks, and 150 to 600° F for class C tanks. Indirectly heated tanks shall also be provided with a thermostatic switch for selection and automatic control of the heat transfer fluid temperature between not more than 100 and 325° F for class A tanks, 100 to 525° F for class B tanks , and 150 to 625° F for class C tanks. Temperature indicating scales shall be marked in increments not greater than 25° F and shall be accurate within +10° F and capable of being recalibrated.

3.6.10 Relay. The electrical control system may utilize a relay for carrying current to the heating units.

3.6.11 Thermometer. A thermometer shall indicate temperature of the dip material. Indirectly heated tanks shall have a thermometer to indicate temperature of the heat transfer fluid. The thermometers shall have a minimum temperature range of 50 to 350° F for class A tanks, 50 to 550° F for class B tanks, and 100 to 650° F for class C tanks. Temperature indicating scales shall be marked in increments not greater than 10° F and shall be accurate within +10° F. If an indicating thermostatic switch is furnished, the thermometer shall not be required.

3.6.12 Heat transfer fluid. Indirectly heated tanks (type III) shall be furnished with a sufficient quantity of heat transfer fluid to permit tank operation. The fluid shall be noncarbonizing and shall have a flash point not less than 620° F.

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3.6.13 Junction box. Unless otherwise specified (see 6.2.1), the hot-dip tank shall be provided with a junction box to permit connection with electric power supply.

3.6.14 Extension cord and plug. When specified (see 6.2.1), portable dip tanks shall be equipped with a 25-foot rubber covered cord with twist lock NEMA standard male plug attached to one end.

3.6.15 Pump. When specified (see 6.2.1), a pump shall be furnished with the tank. The pump shall be electric motor driven and capable of circulating the contents of the dip compartment in not more than 10 minutes for tank capacities not greater than 50 gallons. For the tank capacity greater than 50 gallons, the pump output shall be not less than 5 gallons per minute. The pump shall be readily accessible for servicing and replacement and shall not obstruct the work area inside or outside the dip compartment.

3.6. 15.1 Pump motor. The pump motor shall be totally enclosed, continuous duty, and conform to the requirements of CC-M-1807, with ball bearings. A thermostatic switch shall be installed in the motor circuit to prevent current flow to the motor before the dip material has melted.

3.6. 15.2 Filter. When specified (see 6.2.1), a cleanable type filter shall be provided in the circulating system having 50, 100 or 200 micron filtering capabilities as specified.

3.6.16 Casters. When specified (see 6.2.1), the tank shall be mounted on four, 4-inch or larger diameter steel casters, two swivel type and two rigid type. The casters shall be rated for not less than 50 percent greater than the gross weight of the tank and contents.

3.6.17 Optional equipment. The following optional equipment shall be furnished and shall be fully described (see 6.2.1).

Ventilation hood
Bottom drain (quick opening type)
Automatic fire extinguishing system
Power operated cover

3.6.18 Instruction and information plates. Each tank shall be equipped with instruction plates, including warning and cautions, suitably located describing any special or important procedures to be followed in operating and servicing the equipment, including instructions for satisfactory operation of all the tank controls.

3.6.18.1 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wraparound tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

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3.6.18.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each hot-dip tank and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer s permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.6.19 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.6.20 Workmanship. Workmanship of the hot-dip tank and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspect ion requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all Inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The hot-dip tank shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Dielectric test. The wiring of the electrical system shall withstand a dielectric test voltage of twice the rated voltage plus 1,000 volts rms for a period of one minute applied between the electrical circuits and tank exterior surfaces, Controls shall be operated to ensure testing of all components. The frequency of the test voltage shall be not less than the rated voltage frequency of the tank (see 3.4.3).

4.7.2 Controls and leakage test. This test applies to all types and classes of tanks covered by this specification. The dip compartment of the tank shall be filled with cold water (55° F +20° F) to within two inches of the top of the compartment. The heat transfer jacket of type III tanks shall be filled with heat transfer fluid. The tank thermostat setting shall be 200° F. The tank shall be operated in accordance with the manufacturer's instructions, to start the heat ing cycle to verify proper operation of controls and indicators specified in 3.5.3. A test thermometer shall be used to determine that the liquid temperature has stablized at 200° F +3° F. The tank thermometer and thermostat readings shall not vary by more than 10° F from the test thermometer reading to verify compliance with 3.6.9 and 3.6.11. When

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furnished, the pump shall be operated to determine compliance with 3.6.15. The tank shall be examined for leaks. If no leaks are found, the filled tank shall be allowed to stand undisturbed for 6 hours and then re-examined for leaks.

4.7.3 Performance test. In accordance with the manufacturer's instructions, the tank shall be prepared for operation and filled with heat transfer fluid (see 3.5.). The temperature of the fluid shall be stablized at 70 + 10° F, and with the cover closed, the heating rate of the tank shall be tested for compliance with 3.5. There shall be no evidence of scorching of the fluid. The force required to raise the cover shall be measured and shall not exceed 25 pounds force (3.6.6.1). The ambient temperature around the automatic closing device shall be raised to the point where the device automatically releases. The temperature at which the device releases shall be as specified in 3.6.6.1.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A and level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-E-17555 and MIL-E-16298.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The tank and its associated non-electrical equipment shall be cleaned in accordance with Process C-1 of the drying procedures specified in MIL-P-116. Electric motors and electrical equipment shall be cleaned and dried in accordance with level A requirements of MIL-E-17555 and MIL-E-16298, as applicable.

5.1.1.2 Preservation. The tank and its associated nonelectrical equipment shall be preserved and packaged in accordance with method I of MIL-P-116. Electric motors and electrical equipment shall be preserved and unit packed in accordance with level A requirements of MIL-E-16298 and MIL-E-17555, as applicable. If the equipment supplier determines that a preservative or packaging procedure is required to protect a surface or item, but such procedure is not specified herein, the supplier shall preserve or unit pack the item using a preservative or procedure selected from MIL-P-116.

5.1.2 Level B. Level B preservation shall be the same as level A.

5.1.3 Commercial. Preservation shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing. Packing shall be level A and B or commercial, as specified (see 6.2.1).

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5.2.1 Level A. When weight, size, and design permit, the tank and its components shall be prepared for shipment in its normal operating condition. When disassembly is required the components shall be packed in appropriate containers. The complete tank or disassembled components shall be packed in one or more of the following containers, as applicable, not to exceed the maximum weight limitation shown.

Specification	Weight Limitations
PPP-B-601 Overseas type	1000 lbs.
PPP-B-621, Class 2	1000 lbs.
MIL-C-52950	4000 lbs.
MIL-C-104	30000 lbs.

The arrangement of contents, anchoring, blocking, bracing, cushioning, closure and strapping shall conform to MIL-STD-1186.

5.2.2 Level B. Level B packing shall be the same as level A.

5.2.3 Commercial. Packing shall be in accordance with ASTM D 3951.

5.3 Marking. When level A preservation and packing is required, unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. When commercial preservation and packing is required, unit packs and exterior shipping containers shall be marked in accordance with ASTM D 3951. Additional marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The hot-dip tanks are intended for use in melting, and maintaining in molten state at a prescribed temperature, preservatives, strippable coatings, and sealing compounds. Type III tanks should be used for materials that melt above 250° F. Type I or type III tanks may be used for materials that melt below 250° F. Type II tanks should only be used for heating liquid materials.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class of tank required (see 1.2).

c. Size required (specify length, width and depth) (see 1.2.1).

d. First article, when required (see 3.1).

e. Specify exceptions and additional safety and health requirements (see 3.2.11 and 6.4).

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f. Painting, if different (see 3.3.5).

g. Electrical power supply (specify voltage, hertz, and phase) (see 3.4.1).

h. Power input for tanks larger than 20 cubic feet (specify KW) for class A (see 3.5.1), for class B or C (see 3.5.2).

i. Legs or base supports and height of the top of tank above floor level, if different (see 3.6.1).

j. Overflow drain for tanks with capacity less than 150 gallons, when required (see 3.6.2). (Specify pipe size, location on width or length dimension and dimension from top of tank to bottom of overflow pipe).

k. Cover, if different (see 3.6.6). (All tanks that heat flammable liquids should have an automatic closing cover.)

l. Automatic closing cover, when required (see 3.6.6.1).

m. Programming switch, when required (see 3.6.8).

n. Junction box, if different (see 3.6.13).

o. Cord and plugs, when required (see 3.6.14).

p. Circulating pump and motor, when required (see 3.6.15).

q. Filter, when required specify microns or mesh size (see 3.6.15.2).

r. Casters, when required (see 3.6.16).

s. Optional equipment, when required, fully describe each item or system (see 3.6.17).

t. Nameplate, if different (see 3.6.18.2).

u. First article inspection, if different (see 4.4).

v. Quality conformance inspection, if different (see 4.5).

w. Level of preservation required, different (see 5.1).

x. Level of packing required (see 5.2).

y. Additional marking required (see 5.3).

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6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.11 requires compliance only with those OSHA requirements that concern the hot-dip tank itself. It does not require compliance with those OSHA requirements that concern “the tank in its operating environment” such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual tanks in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirement that will integrate the new tank into its future operating environment. The tank shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer’s commercial tank supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(e).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (keyword) listing:

Hot-dip tank
Directly heated
Indirectly heated
Dip compartment
Heat transfer fluid
Thermometer
Thermostatic switch
Cover, automatic

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA - IP

Project (3426-0038)

Review activities:
DLA - GS
Army ME
Navy - SA
Air Force - 84

User activity:
Army - AT

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-13291F	Tanks, Hot-Dip, Directly and Indirectly Heated

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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